

# A Dual-Tank Model for Real-Time Tracking of Phosphocreatine and Glycolytic Energy Reserves in Cycling

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White paper — AnaerobicFuelTanks project Version 0.3 · 2026-07-11 (second revision — parallel draw with fullness-tapered PCr flux, deficit accounting, and recalibrated recovery)

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## Abstract

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Endurance-cycling analytics have converged on the Critical Power (CP) /  $W'$  framework, in which all work performed above a sustainable power asymptote is drawn from a single finite reserve,  $W'$ . This lumps two physiologically distinct anaerobic systems — the fast phosphocreatine (PCr / alactic) system and the slower glycolytic (lactic) system — into one tank with a single recovery time constant. That simplification is convenient but discards the information a rider most wants during hard, variable-intensity efforts: *which* reserve is being spent, and *how fast* each will come back. This paper proposes a **dual-tank model** that splits  $W'$  into a fast PCr reserve and a slow glycolytic reserve, drives both from the instantaneous power trace, and applies system-specific depletion and restoration laws drawn from the bioenergetic and  $^{31}\text{P}$ -MRS literature. We are explicit up front about what this is: a **physiologically-motivated decomposition of one measured quantity ( $W'$ bal) whose split is assumed, not measured**, and whose extra resolution over single-tank  $W'$ bal is real but concentrated in the transients after hard efforts. It generalizes  $W'$ bal in depletion and diverges from it — by design — in recovery. The model is deliberately reduced to be computable at 1 Hz on a wrist- or bar-mounted device, and we specify a Garmin Connect IQ implementation that displays live reserve, consumption, and depletion for both systems. We close with a validation strategy — including the one out-of-sample test that would actually earn the second tank — and an explicit statement of the model's assumptions and limits.

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## 1. Introduction

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A power meter reports mechanical output, not metabolic cost. The bridge between the two is a model of the body's energy-supply systems. The dominant bridge in cycling — the CP/ $W'$  model — treats supra-CP work as depletion of a single reservoir and sub-CP work as its exponential refill (Skiba et al. 2012). It is power-native, validated, and computationally trivial, which is why it underlies  $W'$ bal displays in GoldenCheetah, intervals.icu, and several head units.

Its central simplification is also its central weakness for real-time decision-making: **W' is one tank**.

Physiologically, anaerobic capacity is at least two tanks with very different dynamics (Margaria; di Prampero & Ferretti 1999):

- The **phosphocreatine (PCr / alactic)** system — high power, low capacity, **fast** oxidative recovery (seconds to a minute), but with recovery that is pH-sensitive and slows across repeated bouts.
- The **glycolytic (lactic)** system — larger capacity, lower peak power, and **slow** recovery (minutes) that effectively only proceeds at low intensity.

A rider deciding whether to cover the next surge cares which tank is empty. If the PCr tank is full but glycolytic is depleted, a 5-second jump is fine but a 40-second dig is not — a distinction the single-tank W'bal cannot express. The goal of this paper is a model that keeps the power-native, on-device virtues of W'bal while restoring the two-system resolution.

Two honest qualifications, stated here rather than buried in §7. First, the split is a **modeling assumption**, not a measurement — power alone cannot identify how W' divides between the two systems (§4.2). Second, the extra resolution is **concentrated in transients**: whenever the fast PCr tank is full — which, given its ~20 s recovery, is most of the time — the glycolytic reserve is just an affine rescaling of the existing single W'bal, so the second bar adds decision-relevant information mainly in the ~30–60 s after a hard effort (which is, however, exactly when covering-the-next-attack decisions are made). The model earns its keep as a *heuristic decomposition*, not as two independently measured reserves.

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## 2. Physiological background

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**Phosphocreatine (alactic).** ATP is buffered by the creatine-kinase reaction ( $\text{PCr} + \text{ADP} \rightarrow \text{ATP} + \text{Cr}$ ). Intramuscular PCr (~20–25 mmol·kg<sup>-1</sup> wet muscle) is the highest-power energy source and is drawn on instantly at any power transient. Its resynthesis is oxidative and fast: classic alactic O<sub>2</sub>-debt half-time ≈ 30 s (Margaria); 31P-MRS gives recovery time constants  $\tau_{\text{PCr}} \approx 20\text{--}40$  s, strongly **pH-dependent** (acidosis slows it) and **biphasic** under high glycolytic load (Harris et al. 1976; Yoshida et al. 2013, PMID 23662804). Recovery also **slows progressively across repeated hard bouts**.

**Glycolytic (lactic).** Anaerobic glycolysis converts glycogen to lactate, supplying large ATP flux for ~10 s–2 min. Its "fuel level" is best tracked as accumulated muscle/blood lactate — a proxy for H<sup>+</sup> and metabolite accumulation. Contribution decays across repeated sprints (≈40% of ATP in the first of 10×6 s efforts, <10% by the last; Sci Rep 2024, DOI 10.1038/s41598-024-78916-z). Restoration is slow (lactic O<sub>2</sub>-debt half-time ≈ 15 min) and, mechanistically, proceeds fastest at low intensity, effectively ceasing above the first lactate threshold (LT1).

**The onset that matters.** Both systems contribute from the start of a hard effort, but not equally at first: PCr is drawn on instantly (it is the highest-power, immediate buffer), while glycolytic flux takes several seconds to ramp up as glycogen phosphorylase is activated (Parolin et al. 1999) — so the *early* demand is PCr-weighted,

shifting toward a shared draw as glycolysis engages. This graded onset — and the very different refill rates — is what the dual-tank model must reproduce (it does so with a parallel, PCr-weighted draw and a glycolytic activation ramp; §4.2), not a strict "PCr first, then glycolytic" hand-off.

**What the two "tanks" represent — and what they don't.** The "tank" is a deliberate simplification for intuition and on-device display, not a claim that either system is a literal reservoir that drains. The two sides are not even the same *kind* of quantity:

- The **PCr tank** does correspond to a real, depletable substrate store — intramuscular phosphocreatine — so "PCr empty" is close to literal.
- The **glycolytic "tank"** is better understood as **tolerance to accumulating fatigue-related metabolites** than as a fuel gauge. On the ~1–4 min timescale it represents, muscle glycogen is nowhere near exhausted; carbohydrate depletion is a separate, hours-long limiter this model does not address. What forces power down at the limit of a hard effort is the buildup of inorganic phosphate (Pi), H<sup>+</sup> (acidosis), ADP, and extracellular K<sup>+</sup> toward the limit of tolerance — consistent with exhaustion at CP/W' coinciding with a reproducible low-PCr / high-metabolite state. (Lactate itself is largely a fuel and a marker, not the cause of force loss.)

This is the physiological reason the two recovery laws differ: PCr resynthesizes quickly, whereas the glycolytic side recovers slowly and only at low intensity because it reflects *clearance and buffering* of accumulated byproducts, which need aerobic metabolism and time — exactly why the model tracks the glycolytic state as an accumulation variable (a muscle-lactate proxy; §4.1) and gates its recovery below LT1. Muscle fatigue is multifactorial and still debated; W' deliberately lumps it into one number for usability. In short: **the PCr tank is a fuel that depletes; the glycolytic "tank" is a byproduct bucket that fills.**

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### 3. Prior models (and why a reduced dual-tank sits between them)

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Three model families exist (full survey in `literature-review-anaerobic-models.md`):

- **Family A — CP / W'-balance** (Skiba 2012/2015; Bartram 2018; Froncioni–Clarke). Power-native and on-device-friendly, but **one lumped tank, single recovery  $\tau$** . Recovery  $\tau_{W'} = 546 \cdot e^{(-0.01 \cdot D_{CP})} + 316 \text{ s}$  cannot represent fast PCr + slow glycolytic simultaneously.
- **Family B — hydraulic tank models** (Morton 1986; Weigend `three_comp_hyd` 2021). **Two explicit anaerobic vessels** (AnF = phosphagen, AnS = glycolytic) plus an aerobic vessel; out-predicts W'bal for intermittent recovery (arXiv 2108.04510). But its 8 parameters are abstract and fit per-athlete by evolutionary computation — heavy for an embedded target.
- **Family C — bioenergetic supply/demand ODEs** (EJAP 2023, PMID 37369795). Explicit alactic / lactic / aerobic metabolic-rate terms, with an efficiency-limited PCr recovery and a power-gated lactate-removal law. Mechanistically ideal but has 17 parameters and needs grey-box fitting.

**The gap:** Family A is implementable but under-resolved; Families B and C resolve the two systems but are too heavy to run and calibrate on a head unit. The dual-tank model below keeps the two-tank resolution of B/C with the on-device simplicity of A.

To be precise about what "lighter" means here, because it is easy to oversell: the model's core parameter count (  $CP$ ,  $W'$ ,  $f_p$ ,  $P_{p\_max}$ ,  $g_{pmax-ratio}$ ,  $\tau_p$ ,  $\tau_g$ ,  $\tau_{on}$ ,  $\tau_{off}$ ,  $LT1$ ,  $\eta$  — **~11**, plus two optional realism terms) is *larger* than the hydraulic model's eight — it is **not** a smaller model. The difference is *where the parameters come from*. Only  $CP$  and  $W'$  are fitted per athlete (from a standard CP test); the rest are **hard-coded to literature defaults** rather than fit by evolutionary computation. That buys on-device simplicity and a trivial calibration, at a real cost: the load-bearing defaults (  $f_p$ ,  $P_{p\_max}$ ,  $LT1$ , and the recovery  $\tau$ 's) are exactly the quantities a CP test does **not** determine, so the model's two-system resolution rests on assumptions, not measurements. That is a legitimate identifiability-for-convenience trade — but it is a trade, and the sections below are explicit about which numbers are earned and which are assumed.

## 4. The dual-tank model

### 4.1 State and parameters

Two energy reserves (Joules), each with a fixed capacity:

- $R_p$  — **PCr reserve**, capacity  $C_p$
- $R_g$  — **glycolytic reserve**, capacity  $C_g$
- Total anaerobic capacity  $W' = C_p + C_g$

The model has **~11 parameters** (plus two optional realism terms). Only  $CP$  and  $W'$  come from a CP test; the rest are literature-fixed defaults (see §3 — this is a real count, not a small one):

| Symbol       | Meaning                                              | Typical default                                              |
|--------------|------------------------------------------------------|--------------------------------------------------------------|
| $CP$         | critical power (W)                                   | from 3-/12-min CP test ( $\approx$ FTP + a few %)            |
| $W'$         | total work above CP (J)                              | from same test ( $\sim$ 15–25 kJ)                            |
| $f_p$        | fraction of $W'$ assigned to the PCr tank            | <b>0.25</b> (assumed; physiological, not from data — see §7) |
| $P_{p\_max}$ | PCr peak power above CP (W), at a <b>full</b> tank   | $\approx P_{1s} - CP$ (best 1 s power – CP)                  |
| $g_{pmax}$   | glycolytic peak power, as a fraction of $P_{p\_max}$ | $0.5 \cdot P_{p\_max}$ (see flux-ratio note)                 |
| $\tau_p$     | PCr recovery time constant (s)                       | 22 (fast, pH-modulated)                                      |
| $\tau_g$     | glycolytic recovery time constant (s)                | <b>520</b> (recalibrated to the reconstitution curve — §6.3) |

| Symbol       | Meaning                                                              | Typical default                                                 |
|--------------|----------------------------------------------------------------------|-----------------------------------------------------------------|
| $\tau_{on}$  | glycolytic activation time constant (s)                              | 6 (Parolin 1999)                                                |
| $\tau_{off}$ | glycolytic <b>de</b> -activation time constant (s)                   | = $\tau_{on}$ (phosphorylase reverts by bout end, Parolin 1999) |
| LT1          | first lactate threshold power (W); glycolytic tank recovers below it | <b>measured</b> (a threshold test), <i>not</i> a fixed %CP      |
| $\eta$       | PCr recovery-rate efficiency (see note)                              | 0.80                                                            |

Derived:  $C_p = f_p \cdot W'$ ,  $C_g = (1 - f_p) \cdot W'$ . Initial state  $R_p = C_p$ ,  $R_g = C_g$  (full).

The  $g_{pmax} = 0.5 \cdot P_{p\_max}$  ratio is not arbitrary: peak glycolytic ATP flux is roughly half peak PCr/creatine-kinase flux in human muscle (di Prampero & Ferretti 1999; the biopsy time-courses of Parolin 1999 / Bogdanis 1996), so PCr is the higher-power system by about 2:1.

Several of these deserve flags up front, because the defaults are doing real work:

- **$f_p$  is assumed, on physiological grounds — not fit to data.** It is *not* determined by a CP test, and (correcting the previous revision) it is *not* pinned by the reconstitution curve either: that curve mainly constrains  $\tau_g$  (§6.3), and the  $f_p$  it implies is confounded with the assumed recovery power of the reference protocol. We set **0.25** from **physiological alactic-fraction estimates** (the PCr system supplies ~20–30% of anaerobic ATP capacity; di Prampero & Ferretti 1999, Bangsbo 1990), and treat it as a per-athlete calibration target, not a measurement.
- **$P_{p\_max} \approx P_{1s} - CP$  is the rate at a FULL tank.** PCr flux is not constant: it **tapers with tank fullness** (creatine-kinase equilibrium — flux falls as PCr depletes), so the available PCr rate is  $P_{p\_max} \cdot (R_p/C_p)$ , equal to  $P_{p\_max}$  only when the tank is full (§4.2). Read  $P_{1s} - CP$  as an upper bound: at 1 s glycolysis is already ~15% active, so it mildly over-attributes to PCr.
- **$\eta$  is a recovery-rate efficiency, not hysteresis.** In the recovery law below it is mathematically equivalent to scaling  $\tau_p$  to  $\tau_p/\eta$  (the asymptote is still full recovery, so there is no path dependence). It is retained only as an interpretable knob mapping to the aerobic-efficiency term of the EJAP 2023 model, and is degenerate with  $\tau_p$  when fitted — the two cannot be identified separately. A *true* efficiency loss (incomplete recovery under fatigue) would require an asymptote **below**  $C_p$ ; that is a possible extension, not what  $\eta$  does here.

## 4.2 The 1 Hz update

Let  $P$  be instantaneous power (W) and  $\Delta t = 1$  s. Define the demand relative to aerobic supply as  $\Delta = P - CP$ .

**Case 1 — depletion ( $\Delta > 0$ , above CP).** Demand  $need = \Delta \cdot \Delta t$  (J) is met by **both systems in parallel**. This matters: glycolytic ATP supply is not instantaneous. Glycogen phosphorylase (the rate-limiting glycolytic enzyme) transforms to its active form over the first several seconds of maximal effort — Parolin *et*

*al.* (1999) measured it rising from 12% at rest to 47% by 6 s — while phosphocreatine is the immediate buffer, near-fully drawn within the first ~10 s (Bogdanis *et al.* 1996); both contribute from the onset, not in sequence (González-Alonso *et al.* 2000). We model this with a **glycolytic activation** term  $g \in [0,1]$  that ramps first-order with a time constant  $\tau_{\text{on}} \approx 6 \text{ s}$  while above CP and relaxes back during recovery:

```
g ← g + (1 - g) · (1 - exp(-Δt/τ_on))      # glycolytic activation ("inertia")

rate_p = P_p_max · (R_p / C_p)              # PCr flux TAPERS with tank fullness (creati
rate_g = 0.5 · P_p_max · g                  # glycolytic peak rate < PCr peak, gated by

# Rate-proportional split: demand shared in proportion to each system's available rate
total_rate = rate_p + rate_g
share_p = need · rate_p / total_rate        # at g=0, full tank → PCr covers all; both
share_g = need - share_p

take_p = min(share_p, R_p, rate_p · Δt)     # both systems rate- AND capacity-limited
take_g = min(share_g, R_g, rate_g · Δt)
unmet = need - take_p - take_g
# shortfall (a tank empty or rate-capped) spills to the partner, then to a deficit:
take_g += min(unmet, R_g - take_g, rate_g · Δt - take_g); unmet -= ...
take_p += min(unmet, R_p - take_p, rate_p · Δt - take_p); unmet -= ...
R_p -= take_p; R_g -= take_g
D += unmet                                # DEFICIT (debt): supra-CP work the caps cou

if unmet > 0: exhaustion = true             # producing power beyond the tanks' rate → a
```

Two structural properties matter, and both were fixed in this revision after they misbehaved when the split and  $f_p$  were changed independently:

- **PCr bottoms out at exhaustion, not before.** With a constant PCr rate cap the tank emptied at a fixed fraction ( $f_p \cdot 1.5 \approx 37\%$ ) of time-to-exhaustion and then read zero for the rest of the effort — contradicting §2's anchor (PCr nadir *coincides* with exhaustion). Tapering the available rate with  $R_p/C_p$  makes  $R_p$  decay asymptotically and reach its minimum at exhaustion, with the split becoming intensity- and history-dependent instead of a flat 2:1. (One extra multiply;  $\text{take}_p + \text{take}_g$  and  $\text{TTE} = W'/\Delta$  are preserved.)
- **Energy is conserved even when a rate cap binds.** If both tanks are rate-limited, the residual  $\text{unmet}$  is measured supra-CP work the rider demonstrably did; discarding it would make combined  $W'$ bal read optimistically high (the dangerous direction). Banking it in a **deficit**  $D$  (which standard  $W'$ bal permits as a negative balance) keeps  $(R_p + R_g - D)$  dropping by exactly  $\Delta$  per second, so combined  $W'$ bal stays energy-conserving and §6.2 holds.  $D$  is repaid during recovery with glycolytic kinetics.

**On identifiability (and consistency with §7).** The parallel draw is the physiologically faithful choice for the *live display*, and  $\tau_{\text{on}}$  is a literature-set constant (Parolin 1999), not a fitted one. But we are careful not to over-claim: the depletion split is **not** identifiable from power at all — total draw above CP is all the pedals see, and any ( $f_p$ ,  $g_{\text{pmax}}$ ) that sums to the same total is observationally equivalent. In *principle* the bi-exponential **recovery** of  $W'$  constrains the split (a fast PCr component + a slow glycolytic component); in *practice* that fit is ill-conditioned (§6.3 shows the fast component is invisible at standard reconstitution sampling), so  $f_p$  ends up **assumed and weakly constrained**, not recovered. This is the §7 position, and it is the one to trust — the reserves are a plausible decomposition, not measured latent state.

**Case 2 — restoration ( $\Delta \leq 0$ , at/below CP).** Recovery "headroom" is  $CP - P$ . Each tank refills exponentially toward full; glycolytic refill is gated by intensity.

```
g ← g · exp(-Δt/τ_off)           # glycolytic DE-activation (τ_off =
D ← D · exp(-Δt/τ_g)           # deficit repaid with glycolytic kin

# PCr: fast, oxidative (η rescales the effective rate to τ_p/η – see §4.1 note)
R_p += η · (C_p - R_p) · (1 - exp(-Δt/τ_p))

# Glycolytic: slow, only meaningfully below LT1 (a MEASURED power, not a %CP)
if P < LT1:
    gate = (LT1 - P) / LT1        # 0 at LT1, → 1 toward zero power
    R_g += gate · (C_g - R_g) · (1 - exp(-Δt/τ_g))
```

The  $g$  de-activation is what makes the activation ramp re-fire on each interval of a repeated-bout set (without it,  $g$  would ratchet to 1 on the first surge and every later bout would start at a flat split — the ramp inert for the rest of the ride).  $\tau_{\text{off}} = \tau_{\text{on}}$  by default: phosphorylase reverts toward basal by the end of a 30 s bout (Parolin 1999), so activation and de-activation share a timescale until data says otherwise.

**LT1** is the rider's **first lactate threshold in watts**, from a threshold test — *not* a fixed fraction of CP. LT1 and CP vary independently (LT1 is roughly 65–85% of CP depending on training status), and this gate decides whether the glycolytic tank refills at all during tempo/endurance riding, so getting it wrong there is consequential. Where only CP is known,  $LT1 \approx 0.80 \cdot CP$  is a fallback, but it is the weakest default in the model and should be replaced by a real measurement.

**Optional realism (recommended, off by default):** - *pH-slowed / fatiguing PCr recovery*: scale  $\tau_p$  upward as the glycolytic tank is emptier, e.g.  $\tau_{p\_eff} = \tau_p \cdot (1 + k \cdot (1 - R_g/C_g))$ , capturing the observed slowing across repeated bouts. - *Aerobic ramp*: replace the hard  $CP$  boundary with a first-order aerobic supply  $A(t)$  that rises toward  $\min(P, CP)$  with  $\tau_{aer} \approx 25$  s, and use  $need = (P - A) \cdot \Delta t$ . This reproduces the onset "oxygen deficit" that transiently loads the anaerobic tanks even below CP.

## 4.3 Reported quantities

- **PCr reserve** =  $R_p / C_p$  (0–100%) — how much "punch" is left.
- **Glycolytic reserve** =  $R_g / C_g$  (0–100%) — how much "sustained dig" is left.
- **Consumption rate** (per system,  $W$ ) —  $take_p/\Delta t$ ,  $take_g/\Delta t$ , the live draw on each tank.
- **Combined  $W'_{bal}$**  =  $(R_p + R_g - D)/W'$  — the deficit term keeps this **energy-conserving in depletion** (it drops by exactly  $\Delta \cdot \Delta t$  per second) so it matches a single-tank  $W'_{bal}$  reference on above-CP segments. It intentionally **diverges in recovery** (bi-exponential + LT1 gate; §4.4/§6.2), so it is *depletion-compatible*, not identical everywhere.
- **Depletion / exhaustion flag** — the rider is producing power beyond the tanks' rate/capacity ( $D$  growing), or both reserves are at/near zero.

## 4.4 Why this behaves correctly

- A short, very hard surge draws mostly from  $R_p$  (glycolysis has not yet ramped in), and  $R_p$  visibly refills within ~30–60 s of easy pedaling — matching PCr physiology. A maximal ~10 s sprint empties  $C_p$ , which *falls out* of the rate rule (it isn't tuned in) and matches Bogdanis 1996.
- A sustained supra-CP effort draws from **both** tanks throughout, with  $R_p$  declining smoothly to its nadir at exhaustion (the fullness-taper; §4.2) rather than emptying early and flat-lining, and  $R_g$  carrying the growing share as  $g \rightarrow 1$ .  $R_g$  comes back only over minutes and only below LT1.
- Verified end-to-end on a synthetic battery (constant supra-CP hold; a 6×[10 s hard / 30 s easy] interval set; a supra-rate-cap effort): PCr reaches its minimum at exhaustion, the activation ramp re-fires on every bout, and combined  $W'_{bal}$  is conserved to the Joule when the caps bind.
- **On "generalizing  $W'_{bal}$ " — a careful claim.** In *depletion* the model is a faithful generalization: the combined balance  $(R_p + R_g - D)$  falls by exactly  $\Delta \cdot \Delta t$  per second (whether or not a cap binds, thanks to the deficit term), so the summed draw above CP matches standard  $W'_{bal}$ . In *recovery* it is deliberately **different**: bi-exponential, with an LT1 gate and the  $\eta / \tau_p$  rate — it does **not** reduce to Skiba's single  $\tau_{W'} = 546 \cdot e^{(-0.01 \cdot D_{CP})} + 316$ . Nor does  $f_p \rightarrow 0$  recover standard  $W'_{bal}$ : it collapses to a *single glycolytic tank* with  $\tau_g$  and an LT1 gate, which is still not the incumbent's recovery law. So this is a generalization in **structure and depletion behaviour**, not a strict superset that contains  $W'_{bal}$  as one parameter setting. The divergence is intentional — the whole point is that single-tank recovery is the thing we are trying to improve on.

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## 5. Real-time on-device implementation (Garmin Connect IQ)

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The model's per-second cost is a handful of multiplies and one `exp()` per tank — trivial for a head unit. A companion document, `connectiq-app-spec-and-prompt.md`, gives the full technical specification and a ready-to-use build prompt. Summary of the target design:



- **App type:** a custom full-screen **Data Field** ( `Toybox.WatchUi.DataField` ), which the head unit calls once per second — a natural fit for the  $\Delta t = 1\text{ s}$  update.
- **Input:** `Activity.Info.currentPower` inside `compute(info)`.
- **Configuration:** `CP`, `W'`, `f_p`, `LT1` (a measured threshold, not a %CP default — §4.2), and the recovery constants exposed via Connect IQ app settings ( `Application.Properties` ), so a rider enters them from Garmin Connect.
- **Display:** two vertical bar gauges (PCr and glycolytic reserve) with numeric % and color bands (green → amber → red), plus optional live consumption in `W`.
- **Recording:** write both reserves and consumption to the FIT file via `FitContributor` fields so they sync to Garmin Connect / intervals.icu / Strava for post-ride analysis.
- **Footprint:** a handful of scalar state variables ( `R_p`, `R_g`, the activation `g`, the deficit `D` ), no arrays — well within Connect IQ memory budgets. Per-second cost is still a few multiplies and one `exp()` per tank.

## 6. Validation strategy

The model is a hypothesis; it must be checked against data before its numbers are trusted. The honest difficulty is that the model's *headline feature* — a separate PCr reserve tracked in real cycling — is the hardest thing to validate, so we separate tests that only check the plumbing from the one test that would actually earn the second tank.

1. **Face validity / unit tests** — synthetic power traces (single sprint, repeated sprints, supra-CP hold, sub-CP recovery) should produce the qualitative behaviours in §4.4.
2. **Backward compatibility — depletion only.** The combined balance  $(R_p + R_g - D)$  reproduces a reference `W'bal` implementation (Froncioni–Clarke) on the same trace **in depletion** — it drops by exactly  $\Delta \cdot \Delta t$  per second, and the deficit term `D` preserves that even when a rate cap binds (which is why the accumulator was added; without it the caps leaked energy and this test failed). It will **not** match in recovery, by construction (§4.4), so this test must be scoped to above-CP segments — a test that expected agreement everywhere is one the model is designed to fail.
3. **Reconstitution targets — what the curve actually constrains.** After a full `W'` depletion, combined recovery can be compared to the published curve ( $\approx 37\% / 65\% / 86\%$  at 2 / 6 / 15 min; Chorley & Lamb 2020). Two things follow, and the second corrects the previous revision: (a) **the curve is blind to the fast tank.** A  $\tau_p \approx 22\text{ s}$  process is  $1 - e^{(-120/22)} \approx 99.6\%$  complete by the first (2 min) sample, so this test validates *glycolytic* recovery and the *size* of the PCr offset — **not** PCr recovery kinetics, i.e. not the feature that distinguishes this model from `W'bal`. To constrain  $\tau_p$  you need samples at  $\sim 10/20/30/45/60\text{ s}$  (test 4). (b) **the curve indicts  $\tau_g$ , not  $f_p$ .** Solving each target point for  $\tau_g$  (passive rest,  $f_p = 0.25$ ) gives 688 / 472 / 537 s — all far above the old 360 s default; the  $\sim 234\text{ s}$  half-time cited for the incumbent implies  $\tau_g \approx 578\text{ s}$  independently. A least-squares fit lands at  $\tau_g \approx 520\text{ s}$ , giving 40 / 62 / 87% vs the 37 / 65 / 86% targets (max error  $\sim 4$  pts, versus 8–9 pts at 360 s). So the default  $\tau_g$  was raised  $360 \rightarrow 520$ ;  **$f_p$  was not re-derived**

**from this curve.** The earlier "matching pulls  $f_p$  down" reasoning was confounded: the implied  $f_p$  swings from  $\sim 0.13$  (passive rest) to  $\sim 0.25$  (soft-pedal recovery) depending entirely on the assumed recovery power of the reference protocol, so the curve cannot pin it.  $f_p = 0.25$  stands on **physiological** grounds (§4.1), not reconstitution. (*Caveat to verify: state the exact recovery power the Chorley–Lamb source protocol used — if it is near-passive, the  $f_p$  it implies is  $\sim 0.13$ , and the physiological anchor is doing all the work of justifying 0.25.*)

4. **Fast-recovery kinetics (the missing piece).** The load-bearing test for  $\tau_p$  is a repeated-bout protocol with **early** recovery sampling — e.g. all-out efforts separated by 10/20/30/45/60 s — and the between-bout power recovery correlated with modelled PCr recovery (as Bogdanis 1996 did for PCr). Standard reconstitution data cannot supply this; it must be measured deliberately.
5. **System-specific truth (lab) — and its hard limit.** Blood-lactate kinetics can corroborate the glycolytic tank, but the PCr tank's gold standard, **31P-MRS, is essentially unavailable in real cycling**: the measurement requires the muscle to be inside a magnet, so it exists only for small-muscle knee-extension or immediate post-exercise, not whole-body cycling. The model's headline novelty therefore **cannot be checked against the gold standard in the modality it targets** — a limitation to state plainly, not paper over.
6. **The test that earns the second tank — out-of-sample intermittent tolerance.** Because the model is a near-superset of single-tank in depletion, it can *always* fit at least as well, so "it fits better" is guaranteed and is **not** evidence the compartments are real. The decisive test is a **head-to-head against single-tank W'bal on an out-of-sample, intermittent-effort tolerance prediction** — the same protocol class on which the hydraulic model beat W'bal (arXiv 2108.04510): predict time-to-exhaustion / whether a prescribed interval set is completable, where the dual-tank and single-tank models make *different* predictions, and show the dual-tank wins. Until that test is passed, the second tank is a plausible, physiologically-motivated decomposition — not a validated one.
7. **Field calibration** — fit  $f_p$ ,  $P_{p\_max}$ , and the  $\tau$ 's per athlete to maximal-effort tests (a sprint-then-hold or repeated-bout protocol) rather than trusting defaults.

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## 7. Assumptions and limitations

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- **The PCr/glycolytic split  $f_p$  is assumed on physiological grounds, not fit to data.** Power alone cannot identify the depletion split (§4.2); recovery can in principle, but the fast component is invisible at standard sampling *and* the reconstitution curve constrains  $\tau_g$  rather than  $f_p$  (§6.3), so  $f_p$  is effectively an assumption. We default to **0.25** from physiological alactic-fraction estimates ( $\sim 0.20$ – $0.30$  of anaerobic ATP capacity; di Prampero & Ferretti 1999, Bangsbo 1990) — *not* from the reconstitution curve, whose implied  $f_p$  is confounded with the reference protocol's recovery power. Treat it as a per-athlete calibration target; outputs are sensitive to it.
- **The reserves are a decomposition, not latent state.** The two bars are a physiologically-motivated split of one measured quantity (W'bal), not two independently measured reserves. When the PCr tank is full — which, given  $\tau_p \approx 22$  s, is most of the time except the  $\sim 30$ – $60$  s after a hard effort — the

glycolytic bar is an affine transform of the existing single  $W'_{bal}$  ( $W'_{bal} = f_p + (1-f_p) \cdot R_g/C_g$ ), so it carries no new information there. The PCr bar adds decision-relevant content **only** in those post-surge transients. Those transients *are* where race decisions concentrate (can I cover this attack right after the last one?), which is the case for the display — but the honest framing is a heuristic decomposition whose split is assumed, not "the two numbers a rider races on" as if both were measured.

- **$\eta$  is degenerate with  $\tau_p$**  (§4.1): as written it rescales the recovery rate, not a true efficiency, and the two cannot be fitted separately. Reported  $\eta / \tau_p$  pairs are non-unique.
- **LT1 should be measured, not derived from CP.** The  $0.80 \cdot CP$  fallback will be wrong for many riders (LT1 ranges ~65–85% of CP), and it gates whether the glycolytic tank recovers during tempo, so a bad value materially changes recovery estimates.
- **The headline feature is hard to validate in-modality.** 31P-MRS cannot be run during real cycling (§6.5), and depletion "fit-better" is guaranteed by construction, so the second tank is only earned by an out-of-sample intermittent-tolerance head-to-head vs single-tank (§6.6) — not yet performed here.
- **Fixed time constants** ignore the documented pH-dependence and bout-to-bout slowing of PCr recovery unless the optional fatigue term is enabled.
- **Hard CP boundary** omits the aerobic slow component and onset kinetics unless the optional aerobic-ramp term is enabled; expect over-attribution to anaerobic tanks in long low-intensity recovery (the same failure mode reported for the EJAP 2023 model).
- **Power is whole-body and mechanical**, not muscle-specific; the model cannot see local fatigue, cadence, or fiber-type effects.
- **Not medical or diagnostic.** This is a training/pacing aid; tank levels are estimates, not measurements of muscle chemistry.
- **Parameters drift** with fitness, fatigue, heat, and altitude; periodic re-testing is required.

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## 8. Conclusion

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The single-tank  $W'$  model earned its place by being power-native and light enough to run anywhere, but it answers "how much anaerobic work is left?" without answering "in which system?" The dual-tank model proposed here offers a physiologically-motivated split at negligible computational cost: divide  $W'$  into a fast, rate-limited, fast-recovering PCr tank and a slow, large, slowly-recovering glycolytic tank; drain them **in parallel** from the power trace (PCr-weighted, with glycolysis ramping in); and refill them with system-specific, intensity-gated laws. It generalizes  $W'_{bal}$  in depletion and diverges from it — deliberately — in recovery, it runs at 1 Hz on a Garmin head unit, and it surfaces a two-system decomposition — *punch left and dig left* — that is most informative in the transients where pacing decisions concentrate.

Two honest caveats set the agenda. The split  $f_p$  is **assumed, not measured**, and the display's extra resolution over single-tank  $W'_{bal}$  is real but concentrated in post-surge windows. And the decomposition is not yet **validated**: the tests that only check the plumbing (backward compatibility, reconstitution offset) it can pass, but the one that would earn the second tank — an out-of-sample intermittent-tolerance win over single-

tank  $W_{bal}$  — remains to be run, and the PCr gold standard (31P-MRS) is unavailable in real cycling. The accompanying Connect IQ specification turns the model into a live data field; the open questions are calibration (  $f_p$  ,  $LT1$  , per-athlete  $\tau$  ) and that decisive head-to-head field validation.

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## References

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See `docs/literature-review-anaerobic-models.md` for the full annotated bibliography with DOIs and PMIDs. Primary sources for this paper:

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*Sourcing note:* equations for the incumbent models were cross-checked against PubMed Central full text (PMC7552657, PMC10638188); typeset equations there were image-embedded, so published closed forms were reconstructed from the surviving prose and standard literature. Verify exact coefficients against publisher PDFs before hard-coding.